



HiDream-I1: An Open-Source High-Efficient Image Generative Foundation Model

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Abstract

Recent advancements in image generative foundation models have prioritized quality improvements but often at the cost of increased computational complexity and inference latency. To address this critical trade-off, we introduce **HiDream-I1**, a new open-source image generative foundation model with 17B parameters that achieves state-of-the-art image generation quality within seconds. HiDream-I1 is constructed with a new sparse Diffusion Transformer (DiT) structure. Specifically, it starts with a dual-stream decoupled design of sparse DiT with dynamic Mixture-of-Experts (MoE) architecture, in which two separate encoders are first involved to independently process image and text tokens. Then, a single-stream sparse DiT structure with dynamic MoE architecture is adopted to trigger multi-model interaction for image generation in a cost-efficient manner. To support flexible accessibility with varied model capabilities, we provide HiDream-I1 in three variants: **HiDream-I1-Full**, **HiDream-I1-Dev**, and **HiDream-I1-Fast**. Furthermore, we go beyond the typical text-to-image generation and remould HiDream-I1 with additional image conditions to perform precise instruction-based editing on given images, yielding a new image editing model namely **HiDream-E1**. We have open-sourced all the codes and model weights of HiDream-I1 and HiDream-E1: <https://github.com/HiDream-ai/HiDream-I1> and <https://github.com/HiDream-ai/HiDream-E1>. These models quickly gained strong traction in the community, ranking among the top globally on the Hugging Face Models Trending list within just one week of launch. In under a month, it surpassed 280,000 downloads and has been officially integrated into the Diffusers library. It is now widely adopted by leading community tools and products, including ComfyUI, Recraft, WaveSpeedAI, fal.ai, and Pruna AI — reflecting the model's growing impact across the open-source AI ecosystem.

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1 Introduction

In the expansive landscape of artificial intelligence (AI), Multimodal Generative AI emerges as a pivotal frontier that amalgamates the generative and multimodal technologies. At its core, Generative AI [1–3, 7, 10–12, 16] excels in creating new content, e.g., text, images, videos or audio, by learning from existing data. Recent advancements in image generative models have dramatically transformed the creative landscape, with pioneers such as Stable Diffusion, Midjourney, DALL-E-3, and Imagen pushing the envelope in text-to-image generation. These models have set new benchmarks for visual fidelity and creativity, fueling applications across digital art, entertainment, design, and beyond. Despite their impressive outputs, many state-of-the-art models grapple with increased computational complexity and longer inference times, making real-time, cost-effective deployment a persistent challenge.

To bridge this gap, we present HiDream-I1, a new open-source image generative foundation model with 17 billion parameters that achieves state-of-the-art image quality within seconds. HiDream-I1 is built on a new sparse Diffusion Transformer (DiT) structure, which first uses dual-stream DiT architecture to separately process image and text tokens and then employs single-stream DiT architecture to enable multi-modal interaction. Both dual-stream and single-stream DiT architecture are remoulded with dynamic Mixture-of-Experts (MoE) design that aims to dynamically route data through specialized expert modules based on input characteristics. In this way, the whole sparse DiT structure achieves a remarkable balance between computational efficiency and output image generation quality.

Understanding that user requirements vary widely across different applications, we offer HiDream-I1 in three distinct variants: 1) **HiDream-I1-Full**: The full-scale model with over 50 diffusion steps, designed for scenarios where the utmost image quality is paramount. 2) **HiDream-I1-Dev**: A guidance-distilled version optimized to operate in 28 diffusion steps, offering a well-balanced

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trade-off between quality and computational demand. 3) **HiDream-I1-Fast**: The fastest version, employing only 14 diffusion steps to deliver high-quality image generation in seconds, ideal for real-time applications. Beyond its robust text-to-image synthesis capabilities, HiDream-I1 lays the foundation for interactive image manipulation (i.e., instruction-based image editing). We extend its functionality with **HiDream-E1**, an instruction-based image editing model that integrates additional image conditions to allow users to perform precise modifications through natural language commands.

In summary, HiDream-I1's contributions are multifaceted:

(1) **Excellent Cost-Effectiveness: Sparse Diffusion Transformer Architecture**. The overall sparse DiT architecture elegantly integrates Sparse Mixture-of-Experts (MoE) technology within the DiT framework. Different expert models can learn to handle various types of text inputs, enabling the model to more accurately understand textual content and generate images with diverse styles and subjects. More importantly, Sparse DiT enhances model performance while controlling computational overhead, achieving exceptional cost-effectiveness.

(2) **Powerful Distillation Technology: GAN-powered Diffusion Model Distillation**. By incorporating generative adversarial learning into diffusion model distillation, the approach leverages GANs' ability to nicely capture details and sharpen edges in image generation, which is commonly sacrificed in typical diffusion model distillation. Accordingly, this way not only distills diffusion model but also further enhances the realism and clarity of the generated images, achieving dual optimization of speed and quality.

(3) **Performance Advantage: Leading Benchmark Results in both human preference and semantic alignment**. Currently, our HiDream-I1 achieves state-of-the-art performances on three benchmarks. Firstly, in the HPS benchmark [15], which holistically assesses semantic relevance, image quality, and aesthetics by calculating the Human Preference Score v2.1 (HPSv2.1), HiDream-I1 attains the best performances across various styles (such as animation, concept art, painting, and photography). Additionally, in the GenEval [4] and DPG-Bench [5] benchmarks, both evaluating the semantic relevance between generated images and input text prompts, HiDream-I1 also achieves top semantic relevance, demonstrating strong prompt-following capabilities.

(4) **Exceptional Scalability: Evolving from Image Generation to Image Editing**. Building upon HiDream-I1, we rapidly expanded to the instruction-based image editing model HiDream-E1. It is worthy to note that after combining text-to-image generation with interactive image editing, HiDream-I1 and HiDream-E1 achieve a "speak-and-it-shall-be" effect similar to GPT-4o's image generation and editing.

2 Model Architecture: HiDream-I1

HiDream-I1 is a large-scale image generative foundation model built upon the principles of flow matching [8] and leveraging a new sparse Diffusion Transformer (DiT) architecture. The core architecture, depicted in Figure 1, comprises two main modules: a hybrid text encoding module and a unique DiT backbone incorporating sparse Mixture-of-Experts (MoE). Similar to previous works (e.g., SD3 [3]), HiDream-I1 operates on latent spaces.

2.1 Hybrid Text Encoding

The ability to accurately translate textual prompts into effective conditioning signals is paramount for high-fidelity text-to-image synthesis. Recognizing that different text encoder architectures possess complementary strengths, HiDream-I1 employs a hybrid strategy integrating representations from four distinct sources:

Long-Context CLIP. We utilize extended-context versions of CLIP-L/14 and CLIP-G/14 [18], which can process longer token sequences than standard CLIP models. These provide robust, globally-aware visual grounding embeddings, presumably pooled into a single vector $h_{\text{clip}} \in \mathbb{R}^d$, used for global conditioning via adaptive layer normalization (adaLN).

T5 Encoder. A T5-XXL encoder is incorporated for its proficiency in parsing complex textual structures, yielding a sequence of contextual token embeddings $h_{t5} \in \mathbb{R}^{M \times d}$.

Decoder-only LLM. To capture deep semantic understanding, we leverage a powerful decoder-only large language model (Llama 3.1 8B Instruct). Crucially, features are extracted from multiple intermediate layers of the LLM, $h_{\text{llm}} \in \mathbb{R}^{L \times M \times d}$ (where L is the number of layers tapped, M is the sequence length, and d is the feature dimensionality), preserving fine-grained semantic details often diluted in final-layer outputs.

As illustrated in Figure 1 (a), the sequence embeddings from T5 (h_{t5}) and the selected intermediate LLM layers (h_{llm}) are processed (e.g., via linear projections) and concatenated to form the primary textual conditioning sequence $T \in \mathbb{R}^{N_T \times d}$ fed into the DiT backbone. This multi-source integration provides a rich, comprehensive textual representation guiding the generation process.

2.2 Sparse Diffusion Transformer (DiT)

HiDream-I1 employs a novel DiT architecture on patchified latent representations. It begins with a dual-stream structure for independent modality processing, and subsequently transitions to a single-stream architecture. Notably, sparse Mixture-of-Experts (MoE) processing is integrated into both the initial dual-stream and the later single-stream stages.

Dual-stream DiT Blocks. The initial blocks of the DiT adopt a dual-stream structure, inspired by MMDiT [3] and typical ViT [9], as shown in Figure 1 (b). Latent patch tokens $I_l \in \mathbb{R}^{N_l \times d}$ (derived from the patchified noisy input X_t) and the combined text tokens $T \in \mathbb{R}^{N_T \times d}$ (from the hybrid encoder) are processed in parallel pathways within these blocks. This allows for specialized initial feature extraction for each modality before interaction in the attention mechanism.

Single-stream DiT Blocks. Following a series of dual-stream DiT blocks (L_{dual}), the architecture transitions into a single-stream configuration. At this juncture, the image tokens and text tokens outputted by the final dual-stream layer ($I_{L_{\text{dual}}}, T_{L_{\text{dual}}}$) are concatenated along their sequence dimension. All subsequent transformer blocks ($l \geq L_{\text{dual}}$) then operate directly on this combined sequence. These single-stream blocks, like the preceding dual-stream blocks, incorporate a sparse Mixture-of-Experts structure for their feed-forward computations.

Sparse Mixture-of-Experts (MoE). Both the dual-stream and single-stream DiT blocks in HiDream-I1 utilize a sparse Mixture-of-Experts (MoE) structure, as detailed in Figure 1 (c, d), in place of

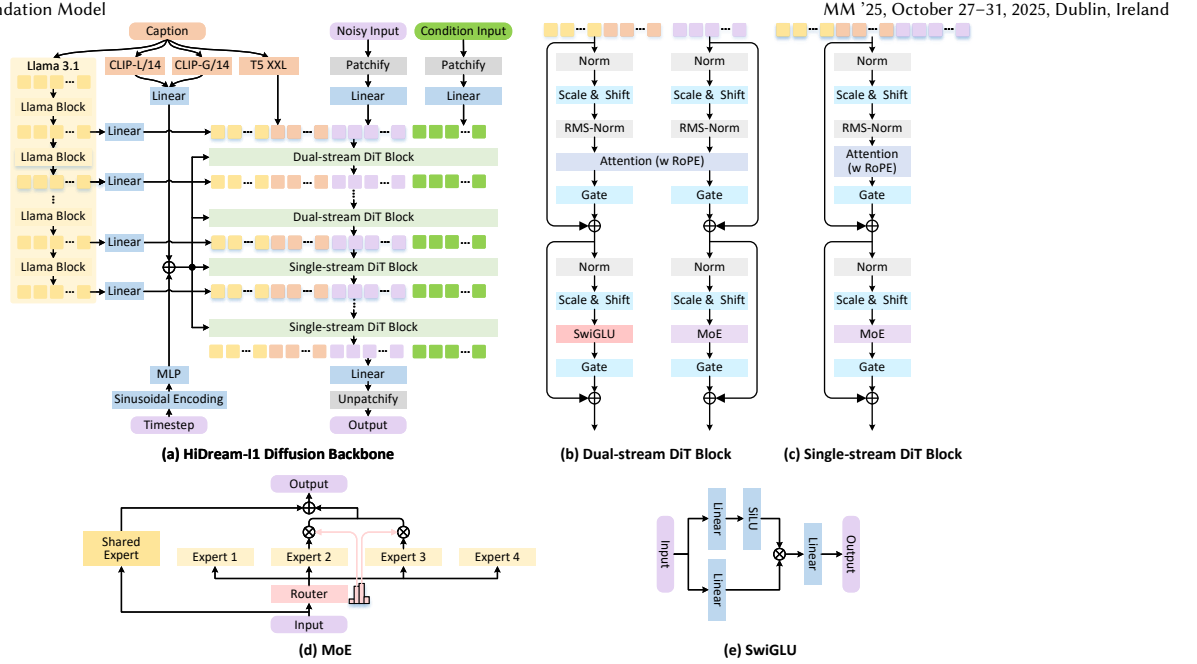


Figure 1: Overall framework of HiDream-I1.

a standard dense feed-forward network (FFN). Within each MoE block, a lightweight gating network, referred to as a router, dynamically directs each input token to a selected small subset of specialized FFN “experts”. As depicted in Figure 1 (d), this includes routing to a designated shared expert as well. The individual expert FFNs employ the SwiGLU [13] (see Figure 1 (e)). This strategy of sparse activation through MoE allows for a significant expansion of the model’s capacity while concurrently maintaining computational efficiency relative to equivalently sized dense models.

Conditioning and Stability. Global conditioning from the pooled Long-CLIP features (h_{clip}) and the sinusoidal timestep embedding are injected into each transformer block. This is achieved via adaptive layer normalization (adaLN), modulating the scale and shift parameters applied to the tokens (the “Scale & Shift” layers in Figure 1 (b, c)). Additionally, QK-normalization [3] is applied within the self-attention mechanisms to enhance training stability.

3 Inference Acceleration via GAN-powered Diffusion Model Distillation

The iterative generation process of the fully trained HiDream-I1 model (denoted HiDream-I1-Full) typically requires approximately 50 sampling steps, potentially limiting real-time applications. To alleviate this issue, we employ knowledge distillation techniques [17] to create significantly faster variants with reduced sampling steps. Specifically, we distill the full model into two accelerated versions: HiDream-I1-Dev, targeting 28 steps, and HiDream-I1-Fast, targeting 16 steps.

The diffusion model distillation process trains the student models (HiDream-I1-Dev and HiDream-I1-Fast) to approximate the generative function of the teacher model (HiDream-I1-Full) with fewer sampling steps. We employ DMD [17] as the primary distillation objective ($\mathcal{L}_{\text{DMD}}$). DMD aims to align the student model’s predicted trajectory distribution with that of the teacher, enabling stable generation with fewer step sizes.

To maintain high perceptual quality and sharpness in the accelerated models, we supplement the DMD loss with an adversarial training objective ($\mathcal{L}_{\text{adv}}$). The student model (generator) is trained concurrently against a discriminator network. This discriminator assesses the realism of images decoded ($\mathcal{D}$) from the student’s generated latents compared to real images. It utilizes multi-level features extracted by the frozen teacher backbone to trigger its classification. The final training objective of this GAN-powered distillation is a weighted sum of the two losses: $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{DMD}} + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}}$.

4 Extension for Image Editing: HiDream-E1

Next, we illustrate how to extend our HiDream-I1 framework to support instruction-based image editing, enabling modifications to a source image (X_S) based on a textual editing instruction (y). Technically, to facilitate editing that is visually grounded in the original image, we leverage an approach that provides strong “in-context” visual conditioning. During fine-tuning, both the source image X_S and the ground-truth target image X_T (the result derived from applying instruction y to X_S) are encoded into the VAE latent space (Z_S, Z_T). These latent maps are spatially concatenated side-by-side. The model is then trained using latent flow matching to generate the target latent Z_T , conditioned jointly on the source latent Z_S (the visual context) and the editing instruction y . To encourage the model to focus learning on relevant modifications dictated by the instruction, we employ a spatially weighted loss function ($\mathcal{L}_{\text{Edit-Weighted}}$). This loss formulation assigns higher weight to errors calculated in latent spatial regions where the target Z_T significantly differs from the source Z_S . This prioritizes accurate generation of the edited areas while helping to preserve the unchanged visual context provided by the source image.

5 Evaluation

To comprehensively assess the generative capabilities of our HiDream-I1 and HiDream-E1, we conducted extensive evaluations against

Table 1: Evaluation results on DPG-Bench.

Model	Overall	Global	Entity	Attribute	Relation	Other
PixArt-alpha	71.11	74.97	79.32	78.60	82.57	76.96
SDXL	74.65	83.27	82.43	80.91	86.76	80.41
DALL-E 3	83.50	90.97	89.61	88.39	90.58	89.83
FLUX.1-dev	83.79	85.80	86.79	89.98	90.04	89.90
SD3-Medium	84.08	87.90	91.01	88.83	80.70	88.68
Janus-Pro-7B	84.19	86.90	88.90	89.40	89.32	89.48
CogView4-6B	85.13	83.85	90.35	91.17	91.14	87.29
HiDream-I1	85.89	76.44	90.22	89.48	93.74	91.83

Table 2: Evaluation results on GenEval.

Model	Overall	Single Obj.	Two Obj.	Counting	Colors	Position	Color Attr.
PixArt-alpha	0.48	0.98	0.50	0.44	0.80	0.08	0.07
SDXL	0.55	0.98	0.74	0.39	0.85	0.15	0.23
FLUX.1-dev	0.66	0.98	0.79	0.73	0.77	0.22	0.45
DALL-E 3	0.67	0.96	0.87	0.47	0.83	0.43	0.45
CogView4-6B	0.73	0.99	0.86	0.66	0.79	0.48	0.58
SD3-Medium	0.74	0.99	0.94	0.72	0.89	0.33	0.60
Janus-Pro-7B	0.80	0.99	0.89	0.59	0.90	0.79	0.66
HiDream-I1	0.83	1.00	0.98	0.79	0.91	0.60	0.72

Table 3: Evaluation results on HPSv2.1 benchmark.

Model	Averaged	Animation	Concept Art	Painting	Photo
Stable Diffusion v2.0	26.38	27.09	26.02	25.68	26.73
Midjourney V6	30.29	32.02	30.29	29.74	29.10
SDXL	30.64	32.84	31.36	30.86	27.48
DALL-E 3	31.44	32.39	31.09	31.18	31.09
SD3	31.53	32.60	31.82	32.06	29.62
Midjourney V5	32.33	34.05	32.47	32.24	30.56
CogView4-6B	32.31	33.23	32.60	32.89	30.52
FLUX.1-dev	32.47	33.87	32.27	32.62	31.11
Stable Cascade	32.95	34.58	33.13	33.29	30.78
HiDream-I1	33.82	35.05	33.74	33.88	32.61

Table 4: Evaluation results on EmuEdit and ReasonEdit.

Model	EmuEdit Benchmark									ReasonEdit Benchmark
	Global	Add	Text	BG	Color	Style	Remove	Local	Average	
MagicBrush	4.06	3.54	0.55	3.26	3.83	2.07	2.70	3.28	2.81	1.75
UltraEdit	5.31	5.19	1.50	4.33	4.50	5.71	2.63	4.58	4.07	2.89
OmniGen	1.37	2.09	2.31	0.66	4.26	2.36	4.73	2.10	2.67	7.36
Gemini-2.0-Flash	4.87	7.71	6.30	5.10	7.30	3.33	5.94	6.29	5.99	6.95
HiDream-E1	5.32	6.98	6.45	5.01	7.57	6.49	5.99	6.35	6.40	7.54

state-of-the-art text-to-image generation models and instruction-based image editing models. We employed several typical benchmarks to measure the text-to-image generation performance of HiDream-I1 in two aspects: 1) prompt adherence on DPG-Bench and GenEval that reflects the model’s ability to accurately understand and execute complex instructions, and 2) human preference on HPSv2.1 benchmark, reflecting the visual appeal of the generated images based on predicted human preference. As shown in Table 1, 2, 3, HiDream-I1 consistently exhibits better performances than other methods across most metrics. For HiDream-E1, we evaluate both the accuracy of the edits made according to instructions and the preservation of image areas not targeted by the edits on EmuEdit [14] and ReasonEdit [6] benchmarks. As shown in Table 4, HiDream-E1 achieves the highest overall average score on EmuEdit and also secures the leading score on ReasonEdit.

6 Conclusion

In this paper, we introduced HiDream-I1, a powerful open-source image generative foundation model that balances state-of-the-art quality with exceptional efficiency. We release three variants of HiDream-I1 (i.e., HiDream-I1-Full, HiDream-I1-Dev, HiDream-I1-Fast) that supports scalable deployment across various use cases, from high-end creative workflows to real-time applications. We further upgrade HiDream-I1 to enable precise instruction-based image editing, leading to a new image editing model HiDream-E1. Experiments conducted on several benchmarks validate the

**Figure 2: Generated images by HiDream-I1.**

superiority of our HiDream-I1 and HiDream-E1 against state-of-the-art approaches.

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